

Jesús Ramírez Nava

Senior Software Engineer · Monterrey, Mexico

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Senior software engineer with 10 years in C/C++ — from 16-bit microcontrollers and CAN bus tooling at John Deere to Amazon Fire TV middleware and API backends today. I've led teams of 10+ engineers, served as Scrum Master, and designed the simulators, test tooling, and backends other teams depend on. Based in Monterrey (UTC-6) and open to remote work with US teams.

EXPERIENCE

Senior Software Engineer · Svitla Systems

Mar 2025 – Present

Project: API backend abstraction to migrate a legal reporting system to a newer platform version

- _ Designing C++ features for an API backend abstraction layer that lets a legal reporting system migrate to its new platform version without breaking existing clients.
- _ Defining clean interfaces between the legacy C++ core and the new API so features can move over incrementally.
- _ Implementing new features and fixing defects through root cause analysis, working with beta users and QA.
- _ Integrating AI-assisted features into the backend and tuning them for response time and cost.

C++ API design Software architecture React AI integration

Senior Software Engineer · GlobalLogic

Aug 2023 – Feb 2025

Project: Amazon Fire TV next generation — porting a third-party video streaming app to new devices and Amazon's new non-Android OS

- _ Ported a major third-party video streaming app to Amazon's next-generation Fire TV devices, working in the middleware layer.
- _ Adapted Google's Cobalt application runtime to the new device hardware and to Amazon's new operating system, which is not based on Android.
- _ Implemented features and fixed defects through root cause analysis, working with beta users and test teams.
- _ Coordinated with partner engineering teams to route and escalate issues outside the project's scope.

C++ Google Cobalt Middleware API/ABI integration Third-party SDK integration
Jira

Senior Software Engineer · GlobalLogic

Aug 2023 – Feb 2025

Project: Test environment to automate embedded-system release processes for an engineering-services provider

- _ Refactored a key desktop application with Python and PySimpleGUI, significantly improving readability and maintainability.
- _ Strengthened the architecture by applying SOLID principles and the listener pattern, producing a more robust, understandable codebase.
- _ Designed and built a project-manager feature from scratch to streamline creating and editing test profiles.

Python PySimpleGUI Git C++ Visual C++ Jira

Software Tech Lead · 3Pillar Global

Apr 2022 – Aug 2023

- _ Analyzed and fixed defects in existing features written in C++ and WinCC OA.

- _ Developed new functionality and unit tests, including keeping system status in sync with the data-queue handshake.
- _ Designed and implemented DevOps infrastructure to support a new licensing model, enabling migration to a modern tech stack.
- _ Built from scratch a Python 3 proof-of-concept API wrapper over crypto.com to simplify requests from higher abstraction layers — the bid was accepted.
- _ Leveled up in Go, Python, API design, and systems design through company training.

C++ WinCC OA Go Python AWS Docker CI/CD Linux Git

Software Supervisor / Technical Lead · John Deere

Feb 2021 – Mar 2022

- _ Led the Mexico team as tech lead and developer, fixing bugs in C++ engine-testing applications.
- _ Supervised 10+ engineers across 5 clients, guiding their professional and personal career paths.
- _ Interviewed candidates and designed team processes to onboard new engineers during headcount growth.

C C++ Qt RTOS Python CI/CD Linux Git

Senior Embedded Software Engineer · John Deere

Dec 2017 – Jan 2021

- _ Scrum Master and developer for embedded applications on 16-bit microcontrollers (C) and their GUIs (Qt C++).
- _ Led the Mexico Qt C++ team troubleshooting the tractor UI — from microcontroller buffer logic to data interpretation and on-screen visualization.
- _ Stood up and led a new team building simulators for the 16-bit microcontroller, the CAN bus, and virtual product interactions in C and C#.
- _ Led design and implementation of a virtual Ethernet simulation supporting Ethernet on 32-bit FreeRTOS microcontrollers.
- _ Reported to users and stakeholders every 2-week sprint and presented the plan for each upcoming PSI.

C C++ C# Qt FreeRTOS CAN Ethernet Python Docker CI/CD Linux

Embedded Software Engineer · John Deere

Dec 2016 – Dec 2017

- _ Developed Qt UI applications, frontend and backend, in C++.
- _ Designed features to process data from a 16-bit microcontroller over CAN (J1939).
- _ Improved code quality by fixing inconsistencies and writing unit tests with Google Test.

C++ Qt CAN J1939 Google Test Jenkins Mercurial Docker Linux

Embedded Software Intern · John Deere

May 2016 – Dec 2016

- _ Designed and built from scratch a static code analyzer with a GUI that highlights bad coding practices, using Flex and Bison.
- _ Built electronic test boxes that emulate signals and communication protocols for device testing.

C C++ CAN Flex Bison HTML CSS

Computer Science Assistant · Tec de Monterrey

Jan 2015 – Aug 2015

- _ Wrote the base code for Kinect applications to improve the classroom experience through technology.
- _ Built Kinect interactions with hardware interfaces and controlled XAML apps through natural body movement in C#.

C# Kinect SDK XAML

EDUCATION

B.Sc. Digital Systems and Robotics Engineering

Tecnológico de Monterrey, Campus Monterrey

2016
MONTERREY, MEXICO

SKILLS

LANGUAGES

C C++ Python C# Go SQL PHP

EMBEDDED

16/32-bit microcontrollers FreeRTOS CAN / J1939 Ethernet Hardware simulation

FRAMEWORKS & UI

Qt React Laravel PySimpleGUI Google Test

BACKEND & SYSTEMS

API design API/ABI integration Third-party SDKs AI backends Systems design

DEVOPS & TOOLS

Docker AWS Jenkins CI/CD Linux Git Jira

LEADERSHIP

Tech lead Scrum Master Team of 10+ Hiring & onboarding

PROJECTS & AWARDS

2019 - 2024	Designed and built a prototype of a next-generation learning management system (LMS) with Laravel and SQL, hosted on DigitalOcean.
2020 - 2021	Designed and built a web app that manages orders for cold-storage spaces, using Laravel and SQL on DigitalOcean.
2016	Built a MISRA-C analyzer that checks C code against the safety-critical coding standard, using Flex and Bison for syntax and semantic analysis.
2015	Built an open-source Android app for programming a robot over Bluetooth with interactive, graphical instructions.
2015	Built a computer-vision system for a Parrot drone that recognized objects around it and responded with matching flight routines.
2015	Parallelized a file-decryption program in C to speed it up.
2015	Created Robython, a high-level programming language and translator that lets children program hardware.
2014	Built a complete data-acquisition system for electrical conductivity: firmware in C plus a C# desktop app that reads the microprocessor and shares data over FTP.
2014	Built an IoT platform that monitors biosignals in real time with humidity, temperature, and heart-rate sensors. Winner, Intel challenge – SISCTI 40
2010	Designed an automatic, programmable pet-food dispenser in C. State winner, DGETI prototypes competition – national finalist

LANGUAGES

Spanish — Native English — B2

CERTIFICATIONS